

Amogh Ravindra Rao

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EDUCATION

Arizona State University

Master's in Data Science, Analytics and Engineering

• **GPA:** 3.77

• **Relevant Coursework:** Statistical Data Analysis, Distributed Database Systems, Data Mining, Big Data Analysis, Data Visualization, Machine Vision and Pattern Recognition, Regression Analysis

RNS Institute of Technology, VTU Board

Bachelor of Engineering in Computer Science

Tempe, AZ

August 2024 – May 2026

Bangalore, India

September 2020 – May 2024

CERTIFICATION AND TECHNICAL SKILLS

Certification: AWS Certified Cloud Practitioner (Link), Ethical Hacking Essential from EC-Council, Data Analysis Job Simulation (Link), GenAI Powered Data Analytics (Link), Quantitative Research Job Simulation (Link)

Skills: Python, SQL (Postgres), R, PyTorch, scikit-learn, Pandas, NumPy, Vision Transformers (ViT), YOLOv8, Power BI, Tableau, FastAPI, Flask, Docker, Git, Azure/AWS/GCP, Agile, Statistical Modeling, A/B Testing, Model Evaluation, Cross-Validation

PROJECTS

Semiconductor Wafer Defect Detection / *Vision Transformer, ResNet, Supervised Contrastive Learning* January 2026 - May 2026

- Partnered with stakeholders to define nine-class defect classification objectives and KPIs; documented current-state processes, target accuracy metrics, and explainability requirements for manufacturing quality control
- Designed a dual-model ensemble combining SmallViT and ResNet-18 trained under a hybrid Cross-Entropy and Supervised Contrastive Learning loss, achieving 0.9749 macro F1 and 97.50% accuracy on 118,595 test samples
- Developed a multi-criteria semi-supervised pseudo-labeling pipeline using five-signal acceptance gating to assign high-quality labels to 638,507 unlabeled wafer maps, achieving 98.30% precision at 75.9% coverage
- Conducted a controlled comparative study between lightweight and full-capacity model configurations, identifying class-imbalance effects and documenting findings in model documentation for governance and retraining decisions

Healthcare Utilization and Diabetes Outcomes / *Tableau Public, Python, Pandas*

August 2025 - December 2025

- Partnered with stakeholders to understand diabetes care objectives and decision processes; translated business problems into 15 interactive Tableau visualizations for administrators, clinicians, and researchers
- Analyzed 101,766 inpatient encounters with 50 demographic and clinical attributes; identified readmission correlations by age, discharge pathway, and medication adjustments to define measurable KPIs
- Implemented interactive filters and dynamic drill-downs with calculated fields to support multi-stakeholder exploration for intervention targeting and policy optimization
- Documented current-state care processes, target outcomes, and acceptance criteria to validate outputs with SMEs and ensure adoption plans were in place

F1 Race Prediction with Feedback Loop / *FastF1, Monte Carlo Dropout*

December 2025 - January 2026

- Built an end-to-end F1 race prediction pipeline by integrating FastF1 telemetry, timing, and results; engineered performance features to model race outcomes with uncertainty quantification
- Trained a deep neural network to predict race times with 95% confidence intervals using Monte Carlo dropout; defined evaluation metrics and monitored model drift
- Implemented a post-race feedback loop that fetches official results, compares predictions vs actuals, tracks error metrics, and stores outcomes for incremental retraining via an experience replay buffer
- Documented functional and non-functional requirements including accuracy targets, latency constraints, retraining triggers, and monitoring plans to ensure operational readiness

LLM Council with Similarity Analysis / *FastAPI, OpenRouter API*

January 2026 - February 2026

- Built a predictive regression model on macroeconomic indicators to detect financial risk patterns and anomalies; defined measurable KPIs including R-squared and RMSE for audit and risk assessment contexts
- Evaluated model accuracy and documented that linear models explained less than 10% of variance, highlighting the need for non-linear modeling approaches and continuous improvement
- Partnered with stakeholders to translate business problems into clear requirements; drafted documentation including process flows, test cases, and release notes for delivery and adoption

Modeling the S&P 500 with Linear Regression / *R, Hypothesis Testing, Regression Analysis*

January 2025 - May 2025

- Built a multi-LLM council system that queried models in parallel and used a chairman model to synthesize higher-quality final responses; defined functional requirements for response quality and evaluation metrics
- Applied sentence embeddings and cosine similarity to evaluate response relevance, detect redundancy, and quantify agreement across model outputs; documented evaluation criteria and acceptance thresholds
- Developed a FastAPI and React interface with streaming stages, heatmaps, relevance charts, and persistent conversation storage for transparent LLM decision-making and UAT validation

EXPERIENCE

Freelance Web Developer

January 2024 – February 2024

AKT Web

Bangalore, India

- Revamped a legacy frontend for a retail client using React.js, delivering a responsive and user-centric interface that enhanced user experience and increased client engagement by 35%
- Improved site performance by implementing lazy loading and other optimization techniques, resulting in a 40% reduction in page load time
- Delivered timely maintenance and support, resolving 80% of reported issues within 24 hours to ensure consistent user experience and system stability